



Optical and X-ray induced luminescence of Sm^{3+} -doped borotellurite and fluoroborotellurite glasses: A comparative study

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ABSTRACT

Borotellurite and fluoroborotellurite glasses doped with Sm^{3+} ions were synthesized and studied their physical, photoluminescence, x-ray induced luminescence, and fluorescence decays. The glasses that were prepared by melt quenching technique were labelled by BTZOS for $30\text{B}_2\text{O}_3 + 29\text{TeO}_2 + 30\text{BaO} + 10\text{ZnO} + 1\text{Sm}_2\text{O}_3$ and BTZFS for $30\text{B}_2\text{O}_3 + 29\text{TeO}_2 + 30\text{BaO} + 10\text{ZnF}_2 + 1\text{Sm}_2\text{O}_3$. The glass becomes more transparent when ZnO was replaced by ZnF_2 . It was found that the replacement of ZnO by ZnF_2 increases absorbance and luminescence of Sm^{3+} ions. The Judd - Ofelt intensity parameters were evaluated from measured absorption spectra for BTZOS and BTZFS glasses and their trend is found to be $\Omega_2 > \Omega_4 > \Omega_6$. The sharp and intense emission peak was originated from ${}^6\text{G}_{5/2} \rightarrow {}^6\text{H}_{7/2}$ transition centred at 600 nm under 403 nm excitation. The colour coordinates of BTZOS and BTZFS glasses are found to fall in the orange region with CCT < 4000 K, suitable for solid-state lighting. BTZFS glass exhibits better radiative properties such as long lifetime, higher quantum efficiency (η), radiative transition probability (A_R), stimulated emission cross-sections (σ_e), branching ratios (β), gain bandwidth and figure of merit, lower non-radiative decay rate (W_{CR}) and saturation intensity (I_s) than BTZOS glass. From the analysis of the results, it is concluded that BTZFS glass could be more suitable for a gain medium for lasers in the visible region.

1. Introduction

Over the past few years, RE^{3+} ion doped optical quality glasses have been more attractive to be investigated due to their sharp emission peaks from ultraviolet to infrared region. The ground state electronic configuration of RE^{3+} is $[\text{Xe}]4f^n$ ($n = 0-14$) and the sharp line is influenced by intraconfigurational f-f transitions [1]. Among RE^{3+} ion, Sm^{3+} ion becomes more attractive one due to its multiple visible emissions with high quantum efficiency [2–4]. Recently, Sm^{3+} ion doped glasses such as Pb–K–Al–Na phosphate [2], TeO_2 – WO_3 – GeO_2 [5], fluorosilicate [6], and borate glasses [7] were widely studied. The results indicate that Sm^{3+} ion doped glasses provide intense emission in 580–680 nm wavelength region when excited by 402–488 nm. The CIE

chromaticity shows that the glasses exhibit colour coordinate lying down at orange to reddish orange region. These emissions could be suitable for photonic applications such as solid-state laser or orange luminescence devices.

Glasses such as silicate, phosphate, tellurite, and borate are suitable materials as a host matrix of the gain medium due to their thermal, mechanical etc., properties. Among these host glasses, borotellurite glass is more interesting to be investigated. Since, borate glass possesses a low melting point, high transparency, high chemical stability, good glass forming nature, and good RE ion solubility [8,9] where as tellurite glass possesses low melting temperature, large refractive index, good chemical and thermal stability [10–12]. Unfortunately, the phonon energy of borate glass is high (~ 1300 – 1500 cm^{-1}) which leads to

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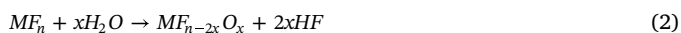
Table 1

Molar compositions of studied glasses along with their mass weight of materials in gram unit, density (ρ , g/cm³), molar volume (V_M , cm³/mol), and refractive index (n) of BTZOS and BTZFS glass.

Composition	Glass code	B ₂ O ₃	TeO ₂	BaO	ZnO	ZnF ₂	Sm ₂ O ₃	ρ	V_M	n
30B ₂ O ₃ + 29TeO ₂ + 30BaO + 10ZnO + 1Sm ₂ O ₃	BTZOS	2.511	5.563	5.529	0.978	–	0.419	3.92	31.83	1.64
30B ₂ O ₃ + 29TeO ₂ + 30BaO + 10ZnF ₂ + 1Sm ₂ O ₃	BTZFS	2.467	5.467	5.433	–	1.221	0.412	3.68	34.50	1.62

strong non-radiative relaxations of RE³⁺ ions in the glass medium. Tellurite glass is a conditional glass-former which possesses fast quenching and needs other glass former to create glass. However, the phonon energy of tellurite glass is low (~ 700 cm⁻¹). Borate and tellurite glass are a good combination to obtain better glass properties such as low phonon energy, high chemical durability, high dielectric constant and excellent transparency [13,14].

For optical gain medium, it is necessary to have long lifetimes for excited states of RE³⁺ ions to achieve high population inversion density, which can be obtained from low electric field strength modifier ions such as Ba²⁺. The addition of BaO in a glass medium increases splitting of absorption and emission peaks due to large ion radius and low local electric field of Ba²⁺ [15]. The addition of heavy metal oxide and/or metal fluorides like ZnF₂ in glass structure is the good choice to decrease non-radiative transition probabilities due to the low phonon energy. Introducing ZnO in borotellurite glass has been studied and the glass medium shows new properties such as decrease of the optical energy gap, increase in refractive index and density of glass medium. ZnO content acts as network modifier at a concentration of 10–20 mol% and a glass forming agent at > 30 mol% [16]. Hence in this research, ZnO plays a role as a network modifier. It is found that the addition of fluoride in the glass matrix can stabilize the vitreous state and possesses a larger transmission range, low refractive index, low dispersion, low Rayleigh scattering, and a high threshold for laser damage [17,18]. The chemical reaction between fluoride and water vapor is most prominent due to the more mobility of fluoride than oxide material. The two possibilities of the chemical reaction between molten glass and water vapor are [19]:



The first reaction occurs in low temperature around T_g , while the second reaction takes place in the high temperature. In the liquid state, the hydroxyl group becomes more active with fluorine anion following chemical reaction (3) which is comparable with (4).



The third chemical reaction is very easy in the liquid state because the hydroxyl group is surrounded by 8–12 fluoride anion. It produces a reduction of OH concentration due to the decrease of neighbouring two OH content in the glass network as the fifth reaction [20].

The effect of oxide/fluoride addition has been studied in glass structures such as Gd₂O₃/GdF₃ in borate glass [21], CaO/CaF₂ in Dy³⁺ ion doped phosphate glass [22], Eu³⁺: TeO₂:B₂O₃:AO:AF₂ (A = Pb, Ba, Zn, Cd and Sr) glasses [14] and ZnO/ZnF₂ in Dy³⁺ ion doped borotellurite glass [23]. The results show that the addition of fluoride produces better luminescence properties that are suitable for solid state and laser applications. However, the addition of oxide/fluoride like ZnO/ZnF₂ in Sm³⁺ ion doped glasses has not been investigated in detail. Hence, it is interesting to study the effect of ZnO/ZnF₂ addition on optical and luminescence properties of Sm³⁺ ion doped borotellurite glass that is developed by melt quenching technique. The physical, optical, luminescence, lifetime and radiative properties of these glasses have been studied and compared to find the best new material for the

photonic application. The spectroscopic properties such as oscillator strength, JO parameters ($\Omega_\lambda = 2, 4, 6$), experimental (β_{exp}) and calculated branching ratios (β_{cal}), radiative transition probability (A_R), and stimulated cross-section (σ_e) have been derived by Judd-Ofelt (JO) theory using measured absorption and luminescence spectra. Moreover, Multipolar interactions, energy transfer mechanisms, X-ray induced luminescence, colour coordinates for these glasses have been reported.

2. Experimental

2.1. Glass preparation

The present glass was prepared by melt quenching technique. The raw materials are powder form including TeO₂ with 99.99% purity from Xijiu, H₃BO₃ with 99.5% purity and ZnO with 99.0% purity from Ajax Finechem Pvt. Ltd., BaCO₃ with 99.0% purity from Himedia Laboratories Pvt. Ltd., ZnF₂ with 99.0% purity and Sm₂O₃ from Sigma-Aldrich have been purchased. The glass composition is presented in Table 1. About 15 g of chemical compositions were mixed in an agate mortar by grinding homogeneously. The raw materials were moved to porcelain crucible, then kept in the electric furnace to melt at 1150 °C for 90 min to get homogeneous melted. The melt was quenched to room temperature by pouring on the rectangular stainless steel plate. For avoiding the glass fracture due to thermal stress, the prepared glass was annealed at 500 °C for about 90 min. Afterwards, the glass can be cooled down in the air, cut them into about 1.5 × 1.0 × 0.3 cm³ dimensions and polished for optical measurements. The glass sample was characterized for physical, optical, photoluminescence, x-ray induced luminescence, radiative properties, and CIE 1931 chromaticity colour coordinates.

2.2. Measurement

The physical properties were characterized by the density (ρ) and molar volume (V_M). Archimedes law was used to obtain the density of the glass sample with water as the immersed liquid. The glass sample was weighed in the air (w_a) and water (w_w) by AND HR-200 balance with ± 0.0001 g accuracy. The formula of density calculation is expressed by Eq. (6). Meanwhile, the formula of molar volume (V_M) was given by Eq. (7).

$$\rho = \left(\frac{w_a}{w_a - w_w} \right) \times \rho_w \quad (6)$$

$$V_M = \frac{M_w}{\rho} \quad ; \quad M_w = x_1 z_1 + x_2 z_2 + x_3 z_3 + \dots + x_n z_n \quad (7)$$

where $x_1, x_2, x_3, \dots, x_n$ and $z_1, z_2, z_3, \dots, z_n$ are mol fraction and molecular weights of compounds, respectively. Refractive index of glass sample was measured by PRESIDUM Refractive Index Meter II. The glass transition, crystallization and melting temperatures of the studied glasses have been determined by Differential Thermal Analyzer (DTA PT1600). About 1 mg of glass sample in powder form was employed for measuring DTA thermogram in temperature range 30–1200 °C with an increase rate of 10 °C/min. The controlled Nitrogen flow rate of 4 L/h was allowed to the furnace. The Energy Dispersive Spectroscopy was undertaken at SEM Laboratory, FMIPA, ITB to detect the constituent elements of the glass sample. The absorption spectra were measured by

Shimadzu 3600 UV-VIS-NIR spectrophotometer where halogen and deuterium lamp are employed as the light sources. The photoluminescence properties including excitation and emission spectra were recorded by Cary Eclipse fluorescence spectrophotometer from Agilent technology Inc. with a Xenon flash lamp as an excitation light source. The absorption, excitation, and emission spectra were measured in 900–1800 nm, 325–575 nm, and 525–750 nm regions, respectively. The fluorescence spectrophotometer was also used to obtain the experimental lifetime for the $^4G_{5/2}$ excited state of Sm^{3+} ions in the studied glasses.

3. Results and discussion

3.1. Physical properties

Molar compositions of studied glasses along with their mass weight of materials in gram unit, density (ρ), molar volume (V_M) and refractive index (n) are presented in Table 1. The density and molar volume of these glasses have been determined using Eqs. (6) and (7), respectively. As can be seen in Table 1, the density and the refractive index of BTZOS glass are higher than BTZFS glass. El-Mallawany observed that the refractive index is increased linearly with increases of density. The lower refractive index of BTZFS glass is due to low polarizabilities and ionic refractivities of fluorine ion [24]. Moreover, the molar volume of BTZFS is higher than BTZOS. The increase of the molar volume of the glass when ZnF_2 replaces ZnO is due to two fluorine atoms of ZnF_2 occupy more space than an oxygen atom of ZnO [22]. It is also observed that BTZFS glass is more transparent than BTZOS glass due to the addition of fluoride atom. It is well known that oxyfluoride glass is hundred times ultra-transparency material than the silicate glass [18].

Basically, the addition of oxide network modifiers to the host matrix improves the glass structure. Addition of network modifier in binary tellurite glass changes the glass structure into two types. The first type is the formation of TeO_4 trigonal bipyramid with an NBO atom. The second type is the transformation of TeO_4 to TeO_3 unit, where the oxygen atom is BO, NBO charged or NBO neutral doubly that are binding with tellurium atom [25]. In the case of TeO_2 - ZnO , the addition of ZnO increases TeO_3 unit [26]. Tellurium atom coordination changes from TeO_4 through TeO_{3+1} polyhedra to TeO_3 and increases the NBO intensity [27]. Meanwhile, the addition of network modifier in borate glass results in two possibilities including NBO formation or conversion of B3 to B4 unit [28]. Fig. 1 shows the FTIR spectra of borotellurite (BTZOS) and fluoroborotellurite (BTZFS) glasses in 650–4000 cm^{-1} region. Several bands are observed and are presented in Table 2. The bands of 671 cm^{-1} in BTZFS glass is Te-O stretching vibration of the trigonal-pyramidal TeO_3 unit [26]. The peaks of 899, 1242 and 1331 cm^{-1} are due to B-O stretching vibrations in BO_4 [29], B-O stretching vibrations in BO_3 from boroxol rings [16] and vibrations of pyroborate groups [14], respectively.

Table 3 shows the EDX data of BTZOS and BTZFS glasses. As can be seen, it is found that there are impurities such as Al and Si in both the glasses. These impurities come from crucible that was used in the melting process. As we know that the constituents of the crucible are Silica > 50% and alumina > 21% [30]. During the process of melt, there is a possible reaction between the glassy melt and crucible material including pouring and solidification process. The reaction between glassy melt and atmosphere in the melting chamber also leads to an increase of impurities. When the glassy melt in the liquid form, there is a strong reaction that increases the homogenous due to stirring effect from melting with high diffusion rate [31]. The Al_2O_3 - SiO_2 system evaporates at a temperature range of 1450–2247 $^{\circ}C$ [32]. Therefore, the applied energy does not enough to remove the impurities in the glass system. From the EDX characterization, Boron and Fluorine element did not detect due to the limitation of the instrument. As we know Boron is difficult to detect together with nitrogen and beryllium due to the low atomic number [33,34]. The light element can produce x-ray with low

energy. However, it is challenging to detect them due to reabsorption of generated x-rays by the sample. The main constituent element of present glasses (Zn, Te, Ba and Sm) have been detected.

The glass transition (T_g), crystallization (T_c) and melting (T_m) temperature of the glass have been determined from the measured DTA curve as shown in Fig. 2. The glass transition is the turnout point in the obstinacy of the baseline, the crystallization temperature can be identified by the beginning of exponential edge from the exothermic process at its T_c and melting temperature is the beginning of the endothermic phase associated with the primary appearance of the fluid stage. They evaluate the glass forming ability (T_g) and thermal stability called Hruby parameter (H_R). Glass forming ability (T_g) is defined by the ratio between T_g and T_m while Hruby parameter (H_R) is expressed by $(T_c - T_g)/(T_m - T_c)$. For a good glass forming, the glass medium should have $T_g \geq 0.67$ and $H_R \sim 0.5$ [35,36]. It can be found that T_g , T_c and T_m of BTZFS glass are 538, 662 and 913 $^{\circ}C$, respectively. Hence, it is appropriate to choose 500 $^{\circ}C$ as annealing temperature since the glass transition temperature is 538 $^{\circ}C$. Meanwhile the T_g and H_R of BTZFS glass are found to be 0.6 and 0.49, respectively. From the DTA analysis, one can conclude that BTZFS glass possesses good glass forming ability and high thermal stability.

3.2. Optical properties

The optical absorption spectra that are observed in 375–1800 nm wavelength region is presented in Fig. 3. The absorbance of the glass is increased with the addition of zinc fluoride in the glass system. However, the replacement of ZnO with ZnF_2 in the glass system does not affect the peak position of optical absorption. The absorption bands are originated from the ground state $^6H_{5/2}$ to several excited levels including $^6P_{3/2}$ (403 nm), $^4I_{13/2} + ^4I_{11/2} + ^4M_{15/2}$ (472 nm), $^4G_{7/2}$ (527 nm), $^6F_{3/2}$ (563 nm), $^4G_{5/2}$ (585 nm), $^6F_{11/2}$ (943 nm), $^6F_{9/2}$ (1080 nm), $^6F_{7/2}$ (1230 nm), $^6F_{5/2}$ (1376 nm), $^6F_{3/2}$ (1477 nm), and $^6H_{15/2} + ^6F_{1/2}$ (1529 nm). The absorption peaks are comparable with the reported data [2,3,37]. The strongest absorption peak arises from the $^6H_{5/2} \rightarrow ^4D_{3/2}$ transition that is called hypersensitive transition centred at 403 nm.

Judd-Ofelt theory [38,39] was used to predict the radiative properties of prepared glasses. The measured parameters are experimental (f_{exp}) and calculated (f_{cal}) oscillator strength, Judd-Ofelt parameter ($\Omega_{\lambda=2,4,6}$), radiative transition probability (A_R), stimulated emission cross section (σ_e), radiative lifetime (τ_r), experimental (β_{exp}) and calculated (β_{cal}) branching ratios. The oscillator strength is derived from absorption spectra, which show the probability of electron transition from ground state to various excited states [40]. The experimental oscillator strength (f_{exp}) is obtained by integrating the molar absorptivity (ϵ) at a wavenumber ν (cm^{-1}) expressed by:

$$f_{exp} = 4.318 \times 10^{-9} \int \epsilon(\nu) d\nu \quad (8)$$

Meanwhile, the calculated oscillator strength is given by:

$$f_{cal} = \frac{8\pi^2 m c \nu}{3h(2J+1)} \frac{(n^2+2)^2}{9n} \times S_{ed} ; \quad S_{ed} = e^2 \sum_{\lambda=2,4,6} \Omega_{\lambda} \langle \Psi J || U^{\lambda} || \Psi' J' \rangle^2 \quad (9)$$

where ' h ' and ' n ' are Planck's constant and refractive indices, respectively, U^{λ} is the doubly reduced matrix element of the unit tensor operator [41], J and J' is a total angular momentum of ground Ψ and upper Ψ' state, respectively. Meanwhile, $(n^2+2)^2/9n$ is Lorentz local field correction. The Judd-Ofelt parameter ($\Omega_{\lambda=2,4,6}$) is related to describe the bonding, symmetry and the rigidity of glass host.

The experimental and calculated oscillator strengths along with JO parameters are presented in Table 4. The hypersensitive transition possesses the highest experimental and calculated oscillator strength. This transition obeys the selection rule $|\Delta J| \ll 2$, $|\Delta L| \ll 2$, $\Delta S = 0$ and very sensitive with surrounding ligand structure. The trend of JO parameters of Sm^{3+} :glasses vary depending on the glass composition

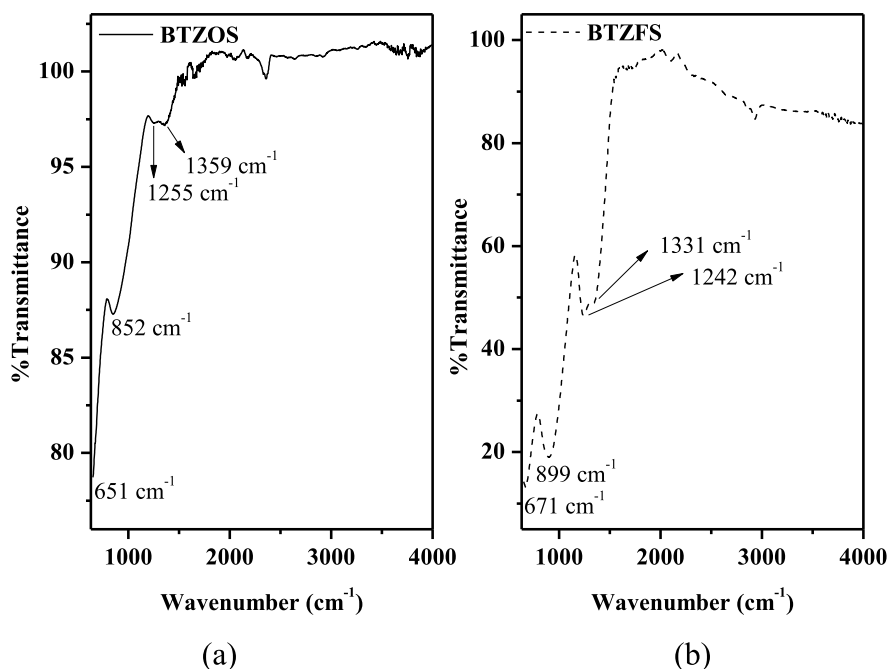


Fig. 1. FTIR spectra of (a) BTZOS and (b) BTZFS glass.

Table 2

Energy of vibrational modes (cm^{-1}) and their assignment in FTIR spectra of studied glasses.

Bands (cm^{-1})		Assignment	Ref
BTZOS	BTZFS		
651	671	Te-O stretching vibration of the trigonal-pyramidal TeO_3 unit	[26]
852	899	B-O stretching vibrations in BO_4	[29]
1255	1242	B-O stretching vibrations in BO_3 from boroxol rings	[16]
1359	1331	Vibrations of pyroborate groups	[14]
1637	1637	H_2O molecular	[57]
2000–3500	2000–3500	Hydrogen bonding and OH stretching	[9,58]

Table 3

Elemental analysis of BTZOS and BTZFS glass using EDX measurement.

Element	Mass%	
	BTZOS	BTZFS
C	25.28	11.19
O	30.13	33.20
Al	1.63	4.85
Si	6.02	7.08
Zn	3.31	3.75
Te	13.45	15.25
Ba	18.03	20.54
Sm	2.14	4.13
Total	100.00	100.00

[2–10,37,42–52]. The PKAPbNSm10 [2], glass C [8], TWGSm10 [5], SNbKZFSm10 [6], SLfSfASm1.0 [37], PABS3 [46], LCaBSm0.3 [48], TZS2 [10], ZPAB [50] glass exhibit the trend of JO parameters as $\Omega_4 > \Omega_6 > \Omega_2$. The PKANbSm10 [3], PKAZLFSm10 [44], ZABBSm0.3% [45], TSZNB [9], BGGS1.00 [47] and 1.5Sm^{3+} :Bismuth silicate [49] glass show the trend of JO parameters as $\Omega_4 > \Omega_2 > \Omega_6$. The JO trendline of BaPbAlFBSm1.0 [4] and PNZSm1 [51] is $\Omega_6 > \Omega_4 > \Omega_2$. The JO trendline of BTZOS and BTZFS glass is

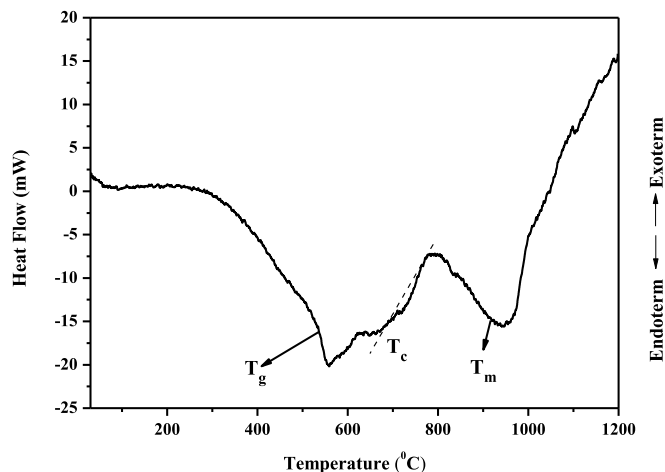


Fig. 2. DTA curve of BTZFS glass.

$\Omega_2 > \Omega_4 > \Omega_6$ and similar to glass C [7], LBGs2 [42] and AECB_Sm1.0 [43]. The JO parameter of Ω_2 shows the asymmetric and covalency between Sm-O bonds [37]. The present glass possesses a higher magnitude of Ω_2 than Ω_4 and Ω_6 indicating lower symmetry in the surrounding Sm^{3+} ion [42].

3.3. Photoluminescence (PL) properties

Fig. 4(a) is the PL excitation spectra that are observed in 325 nm–575 nm wavelength region by monitoring emission at $\lambda_{\text{em}} = 600$ nm. Several excited states such as $^4\text{H}_{9/2}$ (344 nm), $^4\text{D}_{3/2}$ (362 nm), $^4\text{L}_{17/2}$ (374 nm), $^6\text{P}_{3/2}$ (403 nm), $^6\text{P}_{5/2} + ^4\text{P}_{5/2}$ (416 nm), $^4\text{G}_{9/2}$ (439 nm), $^4\text{I}_{13/2} + ^4\text{I}_{11/2} + ^4\text{M}_{15/2}$ (476 nm), $^4\text{G}_{7/2}$ (523 nm) and $^4\text{F}_{3/2}$ (561 nm) have been originated from $^6\text{H}_{5/2}$ ground state [41]. The $^6\text{H}_{5/2} \rightarrow ^6\text{P}_{3/2}$ transition centred at 403 nm is the highest intensity peak of excitation spectra. The replacement of ZnO by ZnF_2 in glass medium increase the intensity of the excitation spectra.

Fig. 4(b) presents the emission spectra of BTZOS and BTZFS glasses measured in the 525 nm–750 nm wavelength region under $\lambda_{\text{ex}} = 403$ nm excitation. Four emission peaks are observed at 562, 600,

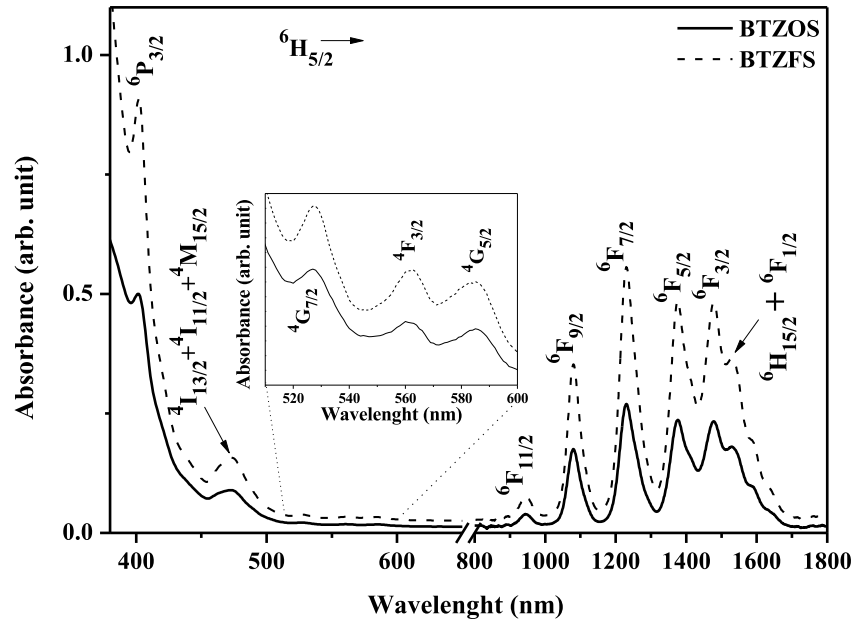


Fig. 3. The optical absorption spectra of BTZOS and BTZFS glasses.

Table 4

Transition, wavelength (λ , nm), experimental energy (E_{exp} , cm^{-1}), experimental ($f_{\text{exp}} \times 10^{-6}$) and calculated ($f_{\text{cal}} \times 10^{-6}$) oscillator strengths of BTZOS and BTZFS glasses.

Transition ${}^4\text{G}_{5/2} \rightarrow$	λ	E_{exp}	BTZOS		BTZFS	
			f_{exp}	f_{cal}	f_{exp}	f_{cal}
${}^4\text{D}_{3/2}$	403	24939	1.10	1.13	1.51	1.98
${}^4\text{I}_{13/2} + {}^4\text{I}_{11/2} + {}^4\text{M}_{15/2}$	472	21186	1.37	1.21	2.90	2.28
${}^4\text{G}_{7/2}$	527	18975	0.04	0.11	0.07	0.21
${}^4\text{F}_{3/2}$	563	17762	0.04	0.01	0.07	0.02
${}^4\text{G}_{5/2}$	585	17094	0.07	0.03	0.12	0.05
${}^6\text{F}_{11/2}$	943	10604	0.57	0.45	0.87	0.85
${}^6\text{F}_{9/2}$	1080	9259	3.04	2.89	5.45	5.42
${}^6\text{F}_{7/2}$	1230	8130	4.83	4.95	9.01	9.06
${}^6\text{F}_{5/2}$	1376	7267	4.42	4.24	7.38	7.42
${}^6\text{F}_{3/2}$	1477	6770	3.54	3.80	6.78	6.64
${}^6\text{H}_{15/2} + {}^6\text{F}_{1/2}$	1529	6540	2.96	2.80	4.79	4.89
RMS			0.139×10^{-6}		0.246×10^{-6}	

644, and 703 nm, which are ascribed to ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{5/2}$, ${}^6\text{H}_{7/2}$, ${}^6\text{H}_{9/2}$, and ${}^6\text{H}_{11/2}$ transitions, respectively [52–54]. The highest intensity of emission peak centred at about 600 nm is due to the ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{7/2}$ transition. The replacement of ZnO by ZnF₂ increase the emission intensity of the glass medium. Due to $\Delta J = 0; \pm 1$, ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{5/2}$ transition is magnetic-dipole allowed, ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{11/2}$ transition is magnetic dipole and electric dipole, ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{5/2}$ and ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{11/2}$ transition are pure electric-dipole allowed. The more intense electric-dipole allowed, the more asymmetric in the host matrices [47].

The partial energy level structure of Sm^{3+} ion in the studied glass is shown in Fig. 5. When the glass sample is excited by 403 nm, the electrons from ground state ${}^6\text{H}_{5/2}$ are moved to several excited states and quickly relaxed to metastable state ${}^4\text{G}_{5/2}$ through multiphonon relaxation and non-radiative cross-relaxation channels as shown in Fig. 5. These processes generate heat and no light production.

Table 5 shows the lasing parameters such as stimulated emission cross-sections, experiment and calculated branching ratios, gain bandwidth, of the present glasses. Radiative transition probability (A_R), stimulated emission cross-sections (σ_e), experiment (β_{exp}) and

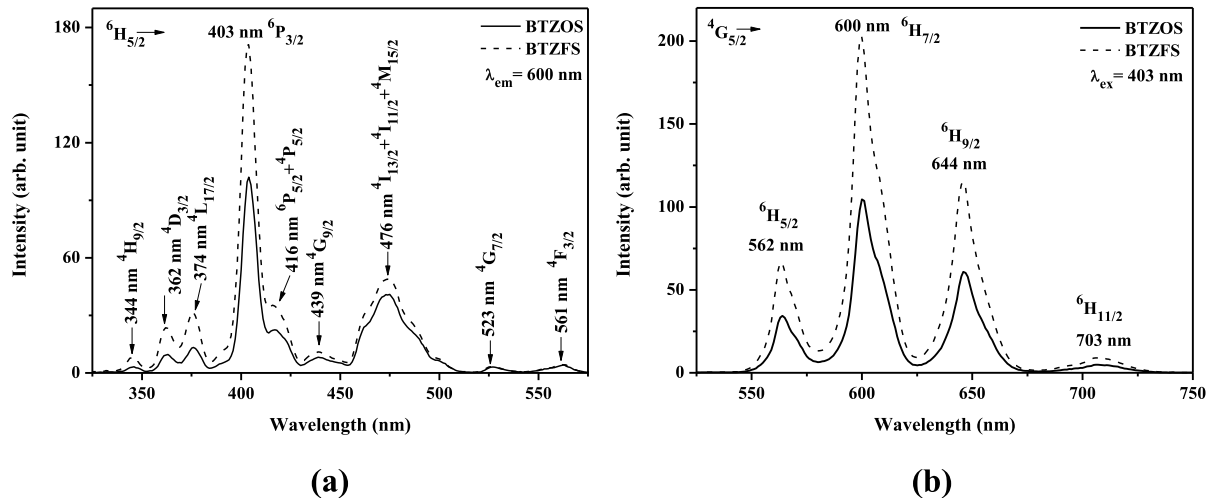


Fig. 4. PL properties of BTZOS and BTZFS glasses: (a) excitation spectra in 300–600 nm region and (b) emission spectra under 403 nm excitation.

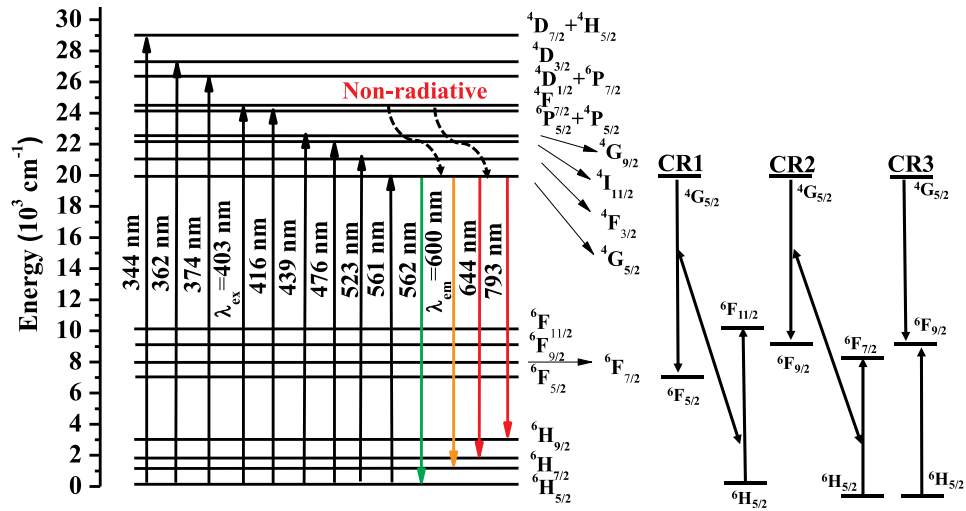


Fig. 5. Partial energy level structure of Sm^{3+} ion in studied glasses. The observed absorption and emission transitions along with probable cross-relaxation channels are shown.

calculated branching ratios (β_{cal}) were calculated using Judd-Ofelt theory [38,39]. Table 6 shows the present glass lasing parameters such as stimulated emission cross-sections, experiment and calculated branching ratios, gain bandwidth, optical gain and figure of merit. Moreover, gain bandwidth is product of emission cross-section (σ_e) and effective linewidth which represents the frequency range where the optical amplification begins. Optical gain is calculated to predict the probability of lasing amplification which is product of emission cross-section (σ_e) and radiative lifetime (τ_{rad}). The figure of merit is an important parameter for laser design system which is determined by multiplying emission cross-section (σ_e) and experiment lifetime (τ_{exp})

[13,37,55]. All this parameters are shown in Table 6. As can be seen, BTZFS glass possesses higher radiative transition probability, stimulated emission cross-sections, experiment and calculated branching ratios, gain bandwidth and figure of merit than BTZOS glass (see Table 6). The replacement of ZnO by ZnF_2 leads to an increase in the value of the lasing parameter. Glass medium saturation intensity (I_s) expressed by Eq. (10) is an important parameter for lasing medium. Saturation intensity corresponds to saturation power that investigates the requirement of initial pump power for achieving saturation.

$$I_s = \frac{hc}{\lambda_e \sigma_e \tau_{exp}} \quad (10)$$

Table 5

JO parameters ($\Omega_\lambda \times 10^{-20} \text{ cm}^2$), their trend, stimulated emission cross-sections ($\sigma_e \times 10^{-21} \text{ cm}^2$), experiment (β_{exp}) and calculated (β_{cal}) branching ratios, efficiency (η , %), experimental lifetimes (τ_{exp} , ms), CIE 1931 chromaticity coordinates, CCT (K), non-radiative decay rate (W_{CR} , s^{-1}), the energy transfer parameter (Q), critical transfer distance (R_0 , Å) and donor-acceptor interaction parameter (C_{DA} , $\times 10^{-41} \text{ cm}^6/\text{s}$) of the Sm^{3+} :glasses.

Glass code	JO parameter			Order	σ_e	β_{exp}	β_{cal}	η	τ_{exp}	(x,y)	CCT (K)	W_{CR}	Q	R_0	C_{DA}
	Ω_2	Ω_4	Ω_6												
BTZOS [pw]	8.13	6.67	3.00	$\Omega_2 > \Omega_4 > \Omega_6$	1.22	52	62	45	0.89	(0.601; 0.399)	1621	617	0.92	5.66	3.69
BTZFS [pw]	14.38	11.86	5.75	$\Omega_2 > \Omega_4 > \Omega_6$	2.04	52	62	81	0.94	(0.599; 0.400)	1622	373	0.93	5.81	4.07
PKAPbNSm10 [2]	2.61	5.87	3.22	$\Omega_4 > \Omega_6 > \Omega_2$	1.15	57	47	84	1.57	(0.509; 0.400)	1626	105	0.802	7.17	5.85
PKANbSm10 [3]	6.89	10.00	2.96	$\Omega_4 > \Omega_2 > \Omega_6$	1.15	42	36	98	1.175	(0.610; 0.370)	—	17.1	1.317	9.20	9.20
Glass C [8]	0.92	2.11	1.29	$\Omega_4 > \Omega_6 > \Omega_2$	0.72	47	44	60	0.966	(0.565; 0.434)	1896	410	2.83	2.83	0.02
TWGSm10 [5]	0.48	4.94	4.64	$\Omega_4 > \Omega_6 > \Omega_2$	0.70	57	50	66	0.79	(0.567; 0.432)	—	432	1.17	9.22	0.53
SNbKZFSm10 [6]	1.19	6.71	2.48	$\Omega_4 > \Omega_6 > \Omega_2$	—	—	—	44	0.79	—	—	707	0.99	7.30	0.12
PKAZLFSm10 [44]	4.50	7.40	3.93	$\Omega_4 > \Omega_2 > \Omega_6$	0.95	56	45	80	1.69	—	—	—	—	—	—
ZABBSm0.3% [45]	4.58	5.62	3.17	$\Omega_4 > \Omega_2 > \Omega_6$	1.23	45	38	80	1.56	(0.593; 0.407)	—	—	—	—	—
SLfSfASm1.0 [37]	2.73	7.49	3.01	$\Omega_4 > \Omega_6 > \Omega_2$	0.92	46	47	34	1.32	—	—	498	1.65	7.52	6.42
Glass C [7]	3.84	2.62	1.69	$\Omega_2 > \Omega_4 > \Omega_6$	1.32	59	41	85	0.734	(0.553; 0.421)	1873	200	2.91	2.71	—
TSZNB [9]	5.21	7.85	3.70	$\Omega_4 > \Omega_2 > \Omega_6$	1.00	40	56	46	0.77	—	—	—	—	—	—
PABNS3 [46]	2.49	4.92	2.56	$\Omega_4 > \Omega_6 > \Omega_2$	0.78	—	54	—	—	(0.588, 0.370)	—	—	—	—	—
BGGs1.00 [47]	1.66	2.89	1.42	$\Omega_4 > \Omega_2 > \Omega_6$	0.48	47	36	28	0.998	(0.602; 0.397)	—	721	0.850	8.74	22.30
LBGS2 [42]	10.72	8.53	7.31	$\Omega_2 > \Omega_4 > \Omega_6$	0.28	36	64	86	0.736	—	—	—	—	—	—
LCaBSm0.3 [48]	1.14	3.32	1.37	$\Omega_4 > \Omega_6 > \Omega_2$	0.93	55	42	58	1.815	(~0.58; ~0.41)	1663	—	—	—	—
TZS2 [10]	0.14	4.42	2.78	$\Omega_4 > \Omega_6 > \Omega_2$	—	—	—	—	—	—	—	—	—	—	—
BaPbAlFBSm1.0 [4]	1.11	3.33	5.61	$\Omega_6 > \Omega_4 > \Omega_2$	1.41	53	52	84	0.750	(0.519 0.467)	—	—	—	—	—
1.5Sm ³⁺ :Bismuth silicate [49]	2.26	3.46	2.09	$\Omega_4 > \Omega_2 > \Omega_6$	0.31	—	49	20	—	(0.610; 0.380)	—	—	—	—	—
ZPAB [50]	2.36	4.84	2.51	$\Omega_4 > \Omega_6 > \Omega_2$	0.83	—	47	—	—	(0.608; 0.383)	—	—	—	—	—
AECSm1.0 [43]	4.33	1.97	1.76	$\Omega_2 > \Omega_4 > \Omega_6$	1.46	56	30	50	1.03	—	—	—	—	—	—
PNZSm1 [51]	6.01	7.27	9.65	$\Omega_6 > \Omega_4 > \Omega_2$	0.47	—	63	79	1.87	—	—	—	—	—	—
BNL1Sm [52]	0.61	1.19	1.18	$\Omega_4 > \Omega_6 > \Omega_2$	0.33	51	40	39	1.24	(0.508; 0.421)	—	490	0.779	6.143	0.003
PKAZLFSm10 [44]	—	—	—	—	—	—	—	—	—	—	—	121	1.36	7.68	0.07
ZABBSm1.0% [45]	—	—	—	—	—	—	—	—	—	—	—	—	1.00	6.07	1.87
PKFBASm10 [53]	—	—	—	—	—	—	—	—	—	—	—	184	0.217	4.85	0.402
PbFPSm10 [54]	—	—	—	—	—	—	—	—	—	—	—	143	0.95	7.55	6.94

Table 6

Transition, wavelength (λ , nm), effective linewidth ($\Delta\lambda_{eff}$, nm), radiative transition probability (A_R , s^{-1}), stimulated emission cross-sections (σ_e , $\times 10^{-21}$ cm²), experiment (β_{exp}) and calculated (β_{cal}) branching ratios, gain bandwidth ($\sigma_e \times \Delta\lambda_{eff}$, $\times 10^{-27}$ cm³), optical gain ($\sigma_e \times \tau_{rad}$, $\times 10^{-24}$ cm²s), figure of merit ($\sigma_e \times \tau_{exp}$, $\times 10^{-24}$ cm²s) of present glasses.

Transition	λ	$\Delta\lambda_{eff}$	A_R	σ_e	β_{exp}	β_{cal}	$\sigma_e \times \Delta\lambda_{eff}$	$\sigma_e \times \tau_{rad}$	$\sigma_e \times \tau_{exp}$
$^4G_{5/2} \rightarrow$									
BTZOS									
$^6H_{5/2}$	562	12.72	70	0.27	0.14	0.05	0.05	0.08	0.03
$^6H_{7/2}$	600	15.51	301	1.22	0.52	0.62	1.90	2.41	1.09
$^6H_{9/2}$	644	16.14	122	0.63	0.31	0.27	1.02	1.24	0.56
$^6H_{11/2}$	703	20.96	15	0.08	0.04	0.06	0.18	0.16	0.07
$A_{tot}(s^{-1})$	508								
$\tau_{rad}(ms)$	1.97								
BTZFS									
$^6H_{5/2}$	562	13.46	106	0.39	0.14	0.05	0.08	0.07	0.06
$^6H_{7/2}$	600	15.83	514	2.04	0.52	0.62	3.23	2.37	1.93
$^6H_{9/2}$	644	16.60	216	1.10	0.31	0.28	1.82	1.28	1.04
$^6H_{11/2}$	703	21.92	26	0.14	0.03	0.06	0.31	0.16	0.13
$A_{tot}(s^{-1})$	861								
$\tau_{rad}(ms)$	1.16								

The saturation intensity (I_s) of BTZOS and BTZFS glass is 31.055×10^8 W/m² and 17.185×10^8 W/m², respectively. From the result, the lower saturation intensity, higher gain bandwidth and figure of merit of BTZFS glass than BTZOS glass indicate that BTZFS glass is more probable for gain medium.

3.4. X-ray induced luminescence

The BTZOS and BTZFS glasses were irradiated with X-rays from Cu target x-ray generator (Inel, XRG3D-E: X-Ray generator) operating at 50 kV and 20 mA for all samples. Luminescence spectra of these irradiated glasses have been measured in 500–1050 nm region under $\lambda_{ex} = 403$ nm excitation wavelength. The spectrometer (QE65 Pro, Ocean Optics) with optical fibre was used to detect emission spectra. The experimental setup of x-ray induced optical luminescence spectrometer is shown in Fig. 6. Fig. 7 exhibits the x-ray induced luminescence spectra measured in 500–1050 nm wavelength region. The spectra shows emission peaks at 564, 601, 645, 710 and 935 nm originated from $^4G_{5/2} \rightarrow ^6H_{5/2}$, $^6H_{7/2}$, $^6H_{9/2}$, $^6H_{11/2}$, and $^6F_{5/2}$ transitions under $\lambda_{ex} = 403$ nm [37,45,55,56]. The transitions of x-ray induced luminescence are identical to the emission spectra with no irradiation except changes in intensity.

3.5. Energy transfer mechanism

The decay curves for the $^4G_{5/2}$ level of Sm^{3+} ions in present glass have been measured by monitoring 600 nm emission corresponds to $^4G_{5/2} \rightarrow ^6H_{7/2}$ transition under 403 nm excitation and are shown in Fig. 8. The experimental lifetime of BTZOS and BTZFS glass is found to be 0.89 and 0.94 ms, respectively. BTZFS glass possesses a higher

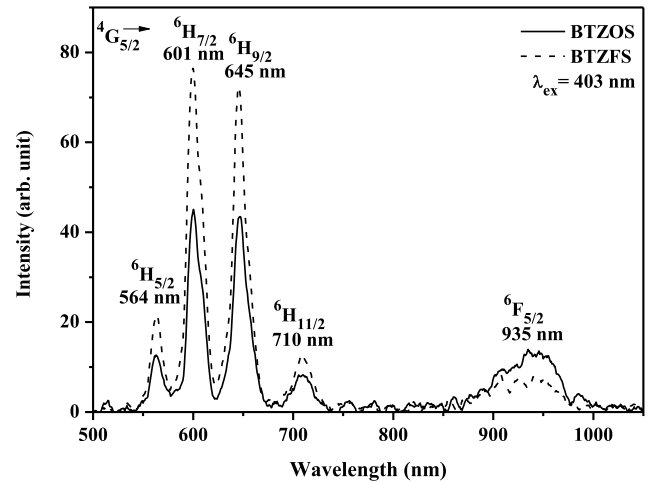


Fig. 7. X-ray induced luminescence of BTZOS and BTZFS glasses.

experimental lifetime (τ_{exp}) than BTZOS glass. The experimental lifetime can be defined by:

$$\frac{1}{\tau_{exp}} = \frac{1}{\tau_{rad}} + W_{MPR} + W_{CR} \quad (11)$$

where W_{MPR} and W_{CR} are the multiphonon and cross-relaxation rate (s^{-1}). The proposed cross relaxation channel of present glass consist of ($^4G_{5/2}$, $^6H_{5/2}$) $\rightarrow$ ($^6F_{5/2}$, $^6F_{11/2}$), ($^4G_{5/2}$, $^6H_{5/2}$) $\rightarrow$ ($^6F_{9/2}$, $^6F_{7/2}$), and ($^4G_{5/2}$, $^6H_{5/2}$) $\rightarrow$ ($^6F_{9/2}$, $^6F_{9/2}$). The quenching of lifetime occurs due to

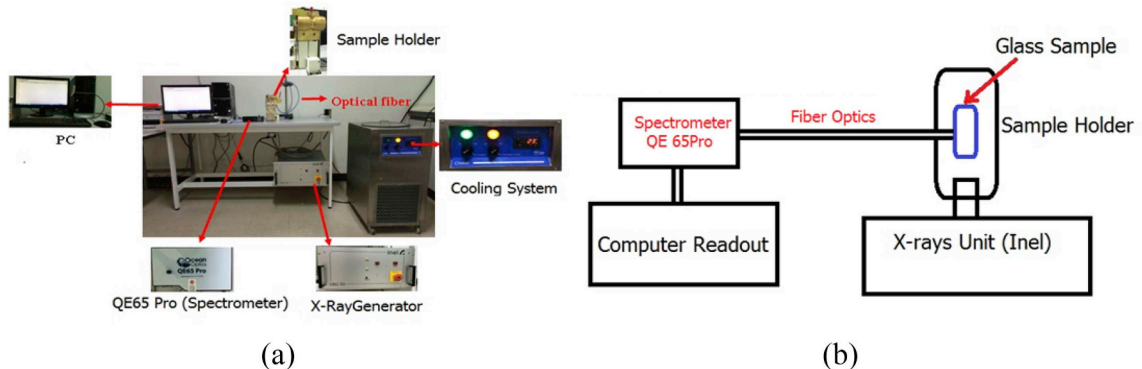


Fig. 6. (a) Photograph and (b) block diagram for measuring the x-ray induced luminescence spectra.

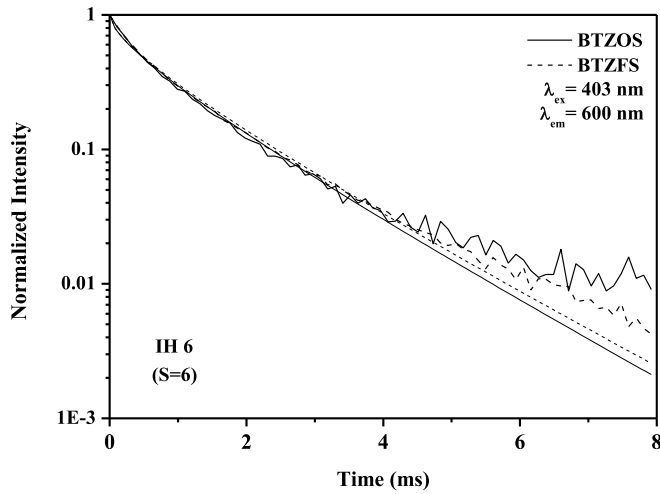


Fig. 8. Decay curve for $^4G_{5/2}$ level of Sm^{3+} ions in BTZOS and BTZFS glasses.

nonradiative transitions including multiphonon relaxation and energy transfer through cross-relaxation. The multiphonon relaxation is negligible due to the high energy gap between metastable $^4G_{5/2}$ level and the next lower $^4F_{11/2}$ level. The cross-relaxation rate W_{CR} of present glass is expressed by:

$$W_{CR} = \frac{1}{\tau_{exp}} - \frac{1}{\tau_R} \quad (12)$$

The efficiency of the present glass can be defined by:

$$\eta = \frac{\tau_{exp}}{\tau_{rad}} \times 100\% \quad (13)$$

The cross-relaxation rate (W_{CR}) and efficiency (η) of BTZOS glass are found to be 617 s^{-1} and 45%, respectively, whereas for BTZFS glass they are found as 373 s^{-1} and 81%, respectively. The W_{CR} and η of BTZOS and BTZFS glass along with other reported Sm^{3+} glasses are presented in Table 5. The W_{CR} of BTZFS glass is lower than BTZOS glass. BTZFS glass have good efficiency and higher than BTZOS, glass C [8], TWGSm10 [5], SNbKZFSm10 [6], PKAZLFSm10 [44], ZABBSm0.3% [45], SLfSfASm1.0 [37], TSZNB [9], BGGS1.00 [47], LCaBSm0.3 [48], 1.5 Sm^{3+} :Bismuth silicate [49], AECB_Sm1.0 [43], PNZSm1 [51] and BNL1Sm [52] glass. These indicate the radiative transition probabilities are more dominant in the BTZFS glass. It is due to the presence of low OH content in BTZFS glass than in BTZOS glass, which decreases the non-radiative transition between Sm^{3+} ions. Therefore, BTZFS glass could be suitable for photonic applications due to the higher efficiency than BTZOS glass.

The measured decay curves for the $^4G_{5/2}$ level of Sm^{3+} ions in present glass can not be fitted to single exponential because donor-acceptor energy transfer notably affects the interaction. The decay curves are fitted to Inokuti Hirayama model which is defined by:

$$I_t = I_0 \exp \left\{ -\frac{t}{\tau_0} - Q \left(\frac{t}{\tau_0} \right)^{\frac{3}{S}} \right\} \quad (14)$$

where I_t is decay intensity at time t , I_0 is initial intensity at time $t = 0$, τ_0 is intrinsic decay time of the donor in absence of the acceptor. S is constant depending on the ion interaction mechanisms such as dipole-dipole interaction ($S = 6$), dipole-quadrupole ($S = 8$) and quadrupole-quadrupole ($S = 10$). Q shows the energy transfer between donor-acceptor that is expressed by:

$$Q = \frac{4\pi}{3} \Gamma \left(1 - \frac{3}{S} \right) N_0 R_0^3 \quad (15)$$

Here $\Gamma(X)$ is 1.77; 1.43, or 1.3 for $S = 6$, 8, or 10, respectively. N_0 is acceptor ions concentration and R_0 is critical energy transfer distance

where the energy transfer rate of acceptor ion is proportional with a decay rate of donor ion. Energy transfer probability (C_{DA}) is given by:

$$C_{DA} = R_0^{(S)} \tau_0^{-1} \quad (16)$$

In this research, the decay curve was fitted to $S = 6$ indicating that the ion interaction in host glass is dipole-dipole interaction. BTZOS glass possesses the value of $Q = 0.919$, $R_0 = 5.659 \text{ \AA}^0$ and $C_{DA} = 3.693 \times 10^{-41} \text{ cm}^6/\text{s}$ whereas BTZFS glass possesses $Q = 0.931$, $R_0 = 5.810 \text{ \AA}^0$, and $C_{DA} = 4.069 \times 10^{-41} \text{ cm}^6/\text{s}$. These parameters for prepared glasses and other reported Sm^{3+} glasses are presented in Table 5. The energy transfer parameter (Q) is influenced not only by Sm^{3+} ion concentration but also glass matrix composition. The Q value of BTZFS glass is lower than PKANbSm10 [3], glass C [8], TWGSm10 [5], SNbKZFSm10 [6], PKAZLFSm10 [44], ZABBSm1.0% [45], SLfSfASm1.0 [37], glass C [7] and higher than BTZOS, PKAPbNSm10 [2], BGGS1.00 [47], BNL1Sm [52], PKFBASm10 [53] glasses. It was found that the critical transfer distance of BTZOS, Glass C [8], Glass C [7] and PKFBASm10 [53] glass is lower than BTZFS glass. The donor-acceptor interaction parameter (C_{DA}) of BTZOS, Glass C [8], TWGSm10 [5], SNbKZFSm10 [6], PKAZLFSm10 [44], ZABBSm1.0% [45], BNL1Sm [52] and PKFBASm10 [53] glasses is lower than BTZFS glass.

3.6. CIE 1931 chromaticity colour coordinates

Fig. 9(a) shows the CIE 1931 chromaticity colour coordinates of BTZOS and BTZFS glasses. The colour coordinates of BTZOS and BTZFS glasses respectively are found to be (0.601; 0.399) and (0.599; 0.400) which fall down at the orange region. The colour coordinate of BTZOS and BTZFS glass along with other reported Sm^{3+} glass is presented in Table 5. The colour coordinates of BTZOS and BTZFS glasses are comparable to reported Sm^{3+} glasses. Fig. 9(b) reveals the light emission of BTZOS and BTZFS glass excited by 408 nm diode laser and it can be seen that the light emission of BTZFS glass is brighter than BTZOS glass. It evidences that the addition of ZnF_2 leads to more intense and stronger emission intensity. The correlated colour temperature (CCT) which is expressed by Eq. (17) illustrates the temperature of a closet Plankin black-body radiator to the operating point on the chromaticity diagram [2]. The CCT describes the light source quality.

$$CCT = -449n^3 + 3525n^2 - 6823n + 5520.3 \quad (17)$$

where $n = (x - x_e)/(y - y_e)$, x and y are colour coordinate from CIE 1931 chromaticity diagram, x_e and y_e are constant corresponding to 0.332 and 0.186, respectively. The CCT of present glass is lower than PKAPbNSm10 [2], Glass C [8], Glass C [7] and LCaBSm0.3 [48] glass. The BTZOS and BTZFS glass possesses CCT value of 1621 K and 1622 K. The CCT value $< 4000 \text{ K}$ belongs to warm light/orange region that can be applied for display devices as well as visible laser application.

4. Conclusion

Sm^{3+} ion doped borotellurite and fluoroborotellurite glasses have been successfully prepared by melt quenching technique. The physical properties show that the replacement of ZnO by ZnF_2 enhance the glass transparency and molar volume. The absorption, excitation, emission and x-ray induced luminescence intensity is increased by introducing ZnF_2 in present glass. The strongest absorption peak arises from the $^6H_{5/2} \rightarrow ^4D_{3/2}$ transition centred at 403 nm. The trends of JO parameters of both BTZOS and BTZFS glasses are found to be $\Omega_2 > \Omega_4 > \Omega_6$ indicating the more covalency nature in the glass structure. The studied glasses under 403 nm excitation show intense orange emission. The photoluminescence spectra under 403 nm excitation are similar to x-ray induced luminescence spectra. BTZFS glass possesses better radiative properties such as higher effective linewidth, radiative transition probability, stimulated emission cross-sections, experimental and calculated branching ratios, gain bandwidth and figure of merit, lower non-radiative decay rate and saturation intensity than BTZOS glass.

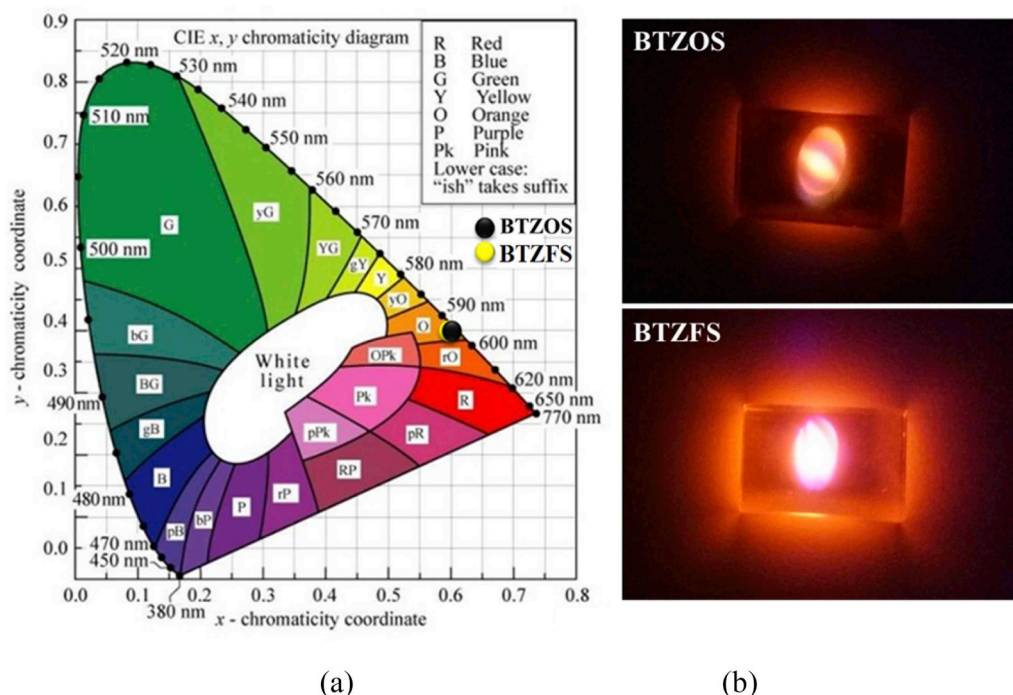


Fig. 9. (a) CIE 1931 chromaticity colour coordinates of BTZOS and BTZFS glasses under $\lambda_{\text{ex}} = 403 \text{ nm}$ and (b) orange light emission of BTZOS and BTZFS glass under excitation of 408 nm diode laser.

Longer experimental lifetime and higher efficiency are obtained for BTZFS glass due to the lower OH content in BTZFS glass. The interaction between optically active Sm^{3+} ions is found to be dipole-dipole in nature and energy transfer between donor-acceptor occur via cross-relaxation. The CCT of present glass is $< 4000 \text{ K}$ that belongs to warm light. All the results indicate that the presence of ZnF_2 can improve the glass properties. These studied glasses especially BTZFS glass could be potential candidates for photonic applications.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jlumin.2019.04.016>.

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